

Electron–phonon and vibronic couplings in the FMO bacteriochlorophyll *a* antenna complex studied by difference fluorescence line narrowing

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Abstract

The electron–phonon and vibronic couplings governing the spectral properties have been studied in the Fenna–Matthews–Olson (FMO) bacteriochlorophyll *a* (BChl *a*)–protein complex at 4.5 K using a spectrally selective difference fluorescence line-narrowing technique. The complex is a part of the light-harvesting system of green photosynthetic bacteria. Its lowest-energy absorption band, peaking at 826 nm and responsible for the fluorescence, is believed to be due to Q_y transitions of largely isolated molecules. One of the main merits of the used method compared with the more common fluorescence line narrowing is that the zero-phonon lines (ZPL) resonant with the excitation laser can be accurately measured, allowing precise determination of the Huang–Rhys (HR) factors, the main characteristics of the linear electron–phonon and vibronic coupling strengths. Over 60 individual vibrational modes of intra- and intermolecular origin have been resolved in the energy range of 45–1600 cm^{−1}. The small HR factors for these modes, ranging between 0.001 and 0.018, add up to a value of $S_{\text{vib}} = 0.38 \pm 0.07$. The effective HR factor for the phonons, S_{ph} , was found clearly wavelength-dependent, varying from ~ 0.7 at short wavelengths to ~ 0.3 at the long-wavelength tail of the absorption spectrum. Coupling between the BChl *a* molecules is likely responsible for this wavelength dependence.

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1. Introduction

Electron–phonon and electron–vibrational (vibronic) couplings are ubiquitous in all material processes where changes of the electronic states occur. Photosynthetic antenna pigment–protein complexes are designed for efficient transfer of energy of the excited electronic states among the pigment molecules within a single complex as well as between the pigments belonging to different complexes. The character of this transport essentially depends on how the pigment electrons are coupled to the dynamic environment. The degrees of freedom of the

environment generally include those of the local intra-pigment vibrations as well as the delocalized phonons of inter-pigment, protein, and solvent origin. It is worth noticing that some of these modes characterize just the coupled guest–host system and do not exist in individual guest or host parts. The common Förster picture emerges when the electron–lattice interaction is strong compared to the coupling between the pigment electrons. The latter coupling as a weak perturbation then induces an excitation energy transfer from a localized state of the energy donor to that of the acceptor. In the opposite case of weak electron–lattice coupling, the zero-order electronic states are delocalized (excitonic), embracing in various extents all the pigment molecules of the antenna complex. Now, the weak perturbation, represented by the electron–lattice coupling, results in energy relaxation down the manifold

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of the exciton states. The electron–phonon and vibronic couplings are thus important characteristics that govern the nature and relaxation dynamics of the electronic states of the antenna complexes. Since the same couplings also shape the absorption and emission spectra of the complexes, a convenient way for their study is at hand [1].

The title antenna complex called the Fenna–Matthews–Olson (FMO) complex is from photosynthetic green sulfur bacteria and contains bacteriochlorophyll *a* (BChl *a*) molecules as the main pigments. It is a trimer of similar protein subunits [2]. Each subunit contains a β -sheet structure with 16 strands that enfolds seven BChl molecules with positions and orientations that are repeated in each subunit. The absorption spectrum of FMO related to the molecular $S_1 \leftarrow S_0$ (Q_y) electronic transition is structured at low temperatures. This structure is mainly assigned to mini-excitons, which account for up to three coupled BChl molecules within the monomer subunit [3]. The lowest-energy 826 nm absorption band responsible for the fluorescence of the FMO complex is, however, believed to be due to three individual transitions in separate BChl molecules, one from each protein subunit [4].

Such quasi-monomer guest molecules are well suited for probing the local guest–host dynamics accompanying the guest electronic transitions, provided the relevant couplings are not too strong. The coupling strengths are commonly described in terms of a dimensionless Huang–Rhys (HR) factor, S (for a review, see Ref. [5]). In the simplest case of a single, harmonic, local mode linearly coupled at low temperature to a valence electron,

$$S = \frac{1}{2} \left(\frac{\Delta q}{\sqrt{\hbar/\mu\omega}} \right)^2. \quad (1)$$

In Eq. (1), Δq is the relative shift between the minima of the potential energy surfaces of the ground and excited states of the electron, $\sqrt{\hbar/\mu\omega}$ is the classical amplitude of the ground state (zero level) local oscillator vibrations, ω is the angular frequency of the oscillator with the reduced mass of μ , and $\hbar = h/2\pi$, where h is the Planck constant. The low-temperature criterion reads $\hbar\omega \gg k_B T$ (k_B is the Boltzmann constant). If this condition is not met, Eq. (1) will be more complex and one should talk about effective HR factor, rather than HR factor.

In the lowest order of the theory comprising adiabatic, harmonic, linear vibronic coupling, and Condon approximations, the conjugate absorption and fluorescence spectra of such a system show mirror symmetry with respect to the pure electronic (or 0–0) line position. The spectra at low temperature, when only the ground vibrational state of the initial electronic state, $|0\rangle$, is populated, consist of a series of narrow lines with intensities distributed according to Franck–Condon factors, $|\langle R|0\rangle|^2 = e^{-S} S^R / R!$ (R runs over the progression of local vibrations in the final electronic state). While the familiar Poissonian distribution, $e^{-S} S^R / R!$, reveals the physical meaning of the HR factor as the mean number of

vibrational quanta created (or lost) in fluorescence (absorption) transitions, the universal relationship $|\langle 1|0\rangle|^2 / |\langle 0|0\rangle|^2 = e^{-S} S / e^{-S} = S$ suggests a practical algorithm for evaluation of S from the experimental spectra.

Avarmaa et al. [6,7] were the first to obtain high-resolution $S_1 \leftrightarrow S_0$ (Q_y) vibronic spectra of impurity BChl molecules in low-temperature glasses. The vibronic line intensities were only measured without the pure electronic origin band that did not allow assessment of vibronic coupling strengths. Currently, two sets of electron–phonon and vibronic HR factors exist, determined for a large number (up to 55) of intramolecular modes of BChl. One series is obtained at 5 K using spectral hole burning (SHB) in a glassy glycerol/water/LDAO matrix [8] and the other [9], applying fluorescence line-narrowing (FLN) method on the FMO antenna complex from the green sulfur bacterium *Prosthecochloris* (*P.*) *aestuarii*. Despite very similar values reported for the electron–phonon coupling strength, S_{ph} , the vibronic HR factors acquired using FLN appear, on average, a factor of three times smaller than those gained from SHB.

According to Ref. [7], the frequencies and relative HR factors of BChl local vibrations measured in fluorescence and fluorescence excitation spectra are very close. Therefore, the question of why the vibronic HR factors determined in Refs. [8,9] differ so badly is important. A closer look on internal limitations of the applied techniques might be worthwhile. In FLN, the zero-phonon line (ZPL) resonant with the laser excitation frequency cannot be measured due to strong elastic light scattering. This is usually not a problem in hole burning where the resonant zero-phonon hole (ZPH) can be easily detected. However, SHB spectroscopy is prone to other systematic errors that might corrupt the HR factors estimated using this method (see, Refs. [10,11] and the discussion in Section 2).

In the present work, a difference version of the FLN technique (further dubbed as Δ FLN) is applied for determination of electron–phonon and vibronic coupling strengths of BChl molecules in the FMO antenna complex. The protein surroundings was favored over that in common solvents because of generally smaller inhomogeneous line broadening in native proteins. As the Δ FLN spectra are virtually free from scattering artifacts [11,12], a more straightforward evaluation of the coupling parameters is expected.

2. Materials and methods

The FMO protein complexes from *Cb. tepidum* were isolated as described in Ref. [2]. The complexes stored in 50 mM Tris–HCl buffer at pH 8.0 were diluted with glycerol (1:2 volume ratio) before placing them into a helium bath cryostat (UTREKS, Ukraine). The samples were contained in cuvettes of 0.6 mm optical path length or in 4 mm diameter gelatin capsules. Optical density of the samples in fluorescence measurements was ~ 0.05 at 826 nm. The sample containers were kept just above the liquid helium bath at a temperature of 4.5 K. The

excitation was provided by a continuous-wave Ti:sapphire solid state laser (Model 3900S, Spectra Physics, USA) with line width of 0.5 cm^{-1} . Detection of the spectra was performed with resolution of 0.1 nm using a 0.6 m spectrograph (MDR-23, LOMO, Russia) in combination with an electrically cooled CCD camera (EEV 30-11, Andor Technology, UK). The fluorescence spectra were corrected for sensitivity of the spectrograph and detector.

As follows, we briefly describe and compare the three relevant line-narrowing techniques: SHB, FLN, and ΔFLN .

SHB [13,14] involves a quasi-monochromatic laser light to selectively bleach a narrow isochromat of pigments within an inhomogeneously broadened absorption band. Hole burning occurs when the resonance with electronic transitions of the pigment is destroyed by photo-induced changes of the pigment, by structural rearrangements of the host environment, or both. The first mechanism is generally referred to as photochemical hole burning, while the second, as non-photochemical hole burning. The non-photochemical hole burning is typical for chromophores in amorphous matrices including inorganic and organic glasses, polymers, and proteins. The SHB spectrum is obtained as the difference between the absorption spectra taken before and after the bleaching process.

It was soon realized that although much more clear-cut than the bands measured by common absorption spectroscopy, the shape of the SHB spectrum does generally not directly represent the homogeneous absorption spectrum, that is, the selectivity of hole burning is limited (see Ref. [15] for reviews). In the case of a single chromophore weakly coupled to the matrix at low temperatures, the homogeneously broadened spectrum consists of series of narrow ZPLs corresponding to the purely electronic origin (0–0 line) and vibronic absorption transitions between the two electronic states of the chromophore. Each ZPL is accompanied from the high-energy side by a broad phonon sideband (PSB) due to coupling of the electronic transition to the matrix phonons. As a result of this, even a monochromatic excitation within the ZPL will not ensure complete spectral selection. A comparably fraction of molecules will simultaneously be excited via their PSB leading to complex hole-burning line shapes.

Fig. 1 schematically pictures the situation for modest electron–phonon coupling along with very weak vibronic coupling. In this case, spectral holes can be burned only around the inhomogeneously broadened purely electronic origin. The SHB spectrum consists of two broad features indicated as real- and pseudo-PSB holes at energies higher and lower compared with the narrow ZPH (at the burn laser frequency), respectively. While the real-PSB indeed represents phonon transitions (of those molecules that are excited resonantly via their ZPLs), the pseudo-PSB only shows the distribution of ZPLs of the molecules that are excited non-resonantly through their PSB. There is also a weaker multi-phonon (multi-PSB) contribution distributed on both sides of ZPH. It represents PSBs of ZPLs

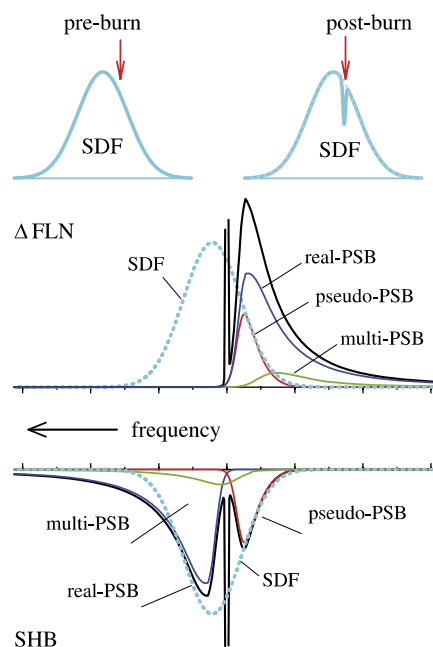


Fig. 1. Model spectra of difference fluorescence line narrowing (ΔFLN) and spectral hole burning (SHB) exposing their component spectra in the 0–0 transition region. The upper part of the figure shows inhomogeneously broadened site distribution functions (SDF) before and after hole burning. The photoproduct spectra are considered lying outside the range of the figure.

associated with the pseudo-PSB holes. The spectral distribution of ZPHs measured at constant non-saturating burn fluence provides the site distribution function (SDF), a measure of static inhomogeneous spectral broadening.

Similarly to SHB in absorption, the FLN spectroscopy employs monochromatic laser light to selectively excite emission from the pigments within their inhomogeneously broadened ensemble [16]. The resulting line-narrowed fluorescence spectrum can again be related to its underlying homogeneous spectrum. SHB and FLN are thus complementary line-narrowing techniques, in the lowest approximations of theoretical models providing equivalent information about the photoactive material centers. Actually, there are differences related how the ZPL and various PSB components are represented in the experimental spectra.

From the first sight, the SHB spectrum, where the real- and pseudo-PSB components lie on opposite sides of ZPH, seems to allow a more straightforward analysis of low-frequency phonon contributions directly from the experiment. In practice, however, the high-energy side of the SHB spectrum is frequently distorted by the absorption of hole-burning products, which in the case of non-photochemical hole burning tend to redistribute randomly within the whole span of SDF. Also, as can be clearly seen from Fig. 1, the limited width of SDF suppresses the high-frequency part of the pseudo-PSB component spectrum. Ignoring these aspects may lead to serious underestimation

of the phonon HR factor, as shown by Pieper et al. [10] and by Rätsep and Freiberg [11]. On the other hand, in FLN the real-, pseudo-, and multi-PSB parts, all lie towards lower energy from the ZPL. Counting pseudo-PSB and multi-PSB as parts of the real-PSB now result in overestimation of the HR factor. Even more serious obstacle in this case is the excitation laser-scattering artifact (Rayleigh scattering) that usually totally obscures the ZPL.

These shortcomings of the FLN and SHB techniques are well recognized. Various solutions to handle the problem, typically exploiting optical saturation of the zero-phonon transitions due to vastly different absorption cross-sections of ZPL and its PSB, have been proposed and analyzed (see reviews by R. Avarmaa and J. Kikas in Ref. [15]). The Δ FLN spectroscopy that uses persistent SHB to gain selectivity was identified among the most promising techniques [15,17–19]. As the difference between the pre- and post-burn absorption spectra yields the SHB spectrum, that between the selective emission spectra produces the Δ FLN spectrum. The Δ FLN spectrum thus only shows the contribution of the emission of the sites or states that were burnt out during irradiation. Despite the high expectations, the early demonstrations of the technique [15,17–19] still failed to record ZPLs resonant with the excitation. The underlying technical difficulties were only recently overcome [12,20] owing to modern technology of simultaneous recording of the whole spectrum by CCD camera and to successful efforts of minimizing scattered laser light.

3. Experimental results

Fig. 2 shows a non-selectively excited fluorescence spectrum of FMO complexes at 4.5 K. The spectrum consists of a single inhomogeneously broadened band with

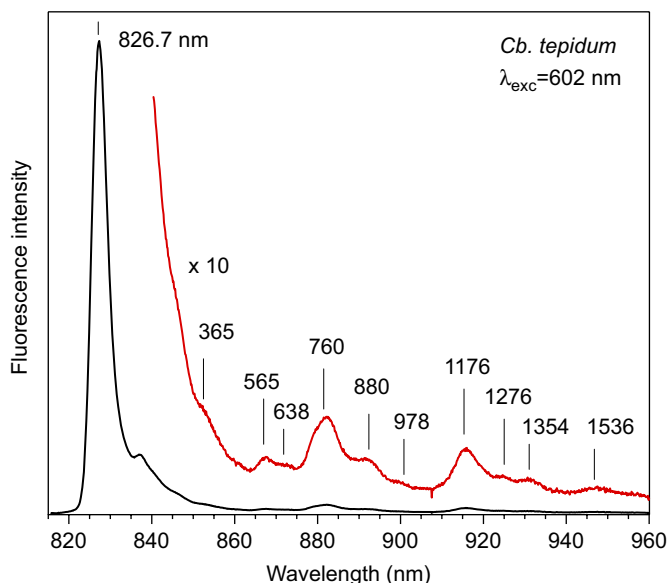


Fig. 2. The 4.5 K non-line-narrowed fluorescence spectrum of the FMO trimer complex excited into the Q_x absorption band at 602 nm. The numbers label vibrational mode frequencies in wave numbers.

full width at half maximum (FWHM) of $72 \pm 5 \text{ cm}^{-1}$. The peak of the fluorescence spectrum at $826.7 \pm 0.2 \text{ nm}$ is almost in resonance with the conjugate absorption maximum at $825.6 \pm 0.2 \text{ nm}$. The main fluorescence band is accompanied from the long-wavelength side by a shoulder at 837 nm and a number of partially resolved satellite features, corresponding to the ground-state vibrations of the BChl molecule. The shoulder can be due to low-energy intramolecular modes, and/or to a residual reaction center core complex (see Section 4).

The vibronic structure of the fluorescence spectrum is more clearly observed upon selective excitation inside the lowest absorption band between 825 and 829 nm, applying the FLN technique. The spectra obtained at excitation wavelengths above 826.8 nm look all the same (Fig. 3), demonstrating the maximum selectivity achievable by this method. Since the 62 vibronic lines identifiable in the spectra are relatively weak, one may be confident that they are due to single-quantum excitations in the S_0 ground state of the BChl molecule. As typical for the FLN spectra, the purely electronic lines are totally obscured by the scattered excitation laser light. This scattering makes quantitative evaluation of the electron–phonon or vibronic coupling strengths, the main purpose of this work, virtually impossible from these data.

We now demonstrate with the help of the Δ FLN technique a possibility of recording the line-narrowed fluorescence spectrum complete with the pure electronic band. Fig. 4 shows series of FLN spectra taken adjacent to the excitation laser wavelength after increasing hole-burning light exposures between successive measurements. The same laser wavelength at 827.1 nm (close to the 826 nm

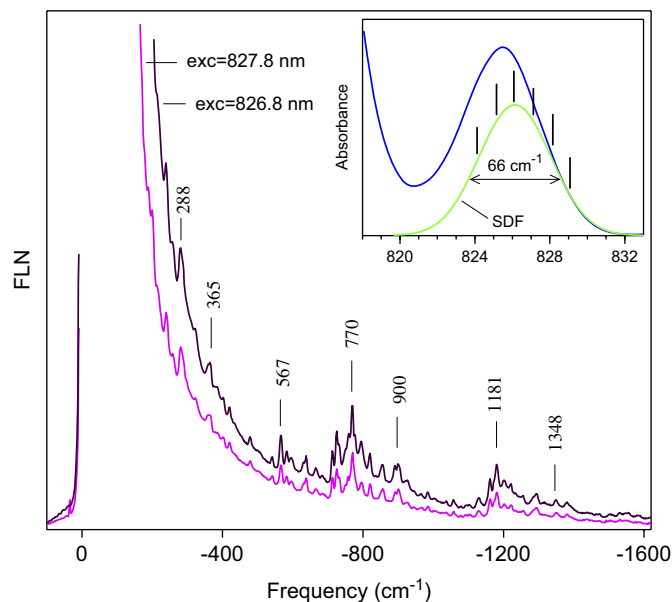


Fig. 3. Two FLN spectra of the FMO complex at 826.8 and 827.8 nm excitations. Low excitation intensity and short illumination time preventing sizeable hole burning was used in those measurements. The inset shows the long-wavelength part of the absorption spectrum of FMO complexes together with SDF determined in Ref. [23].

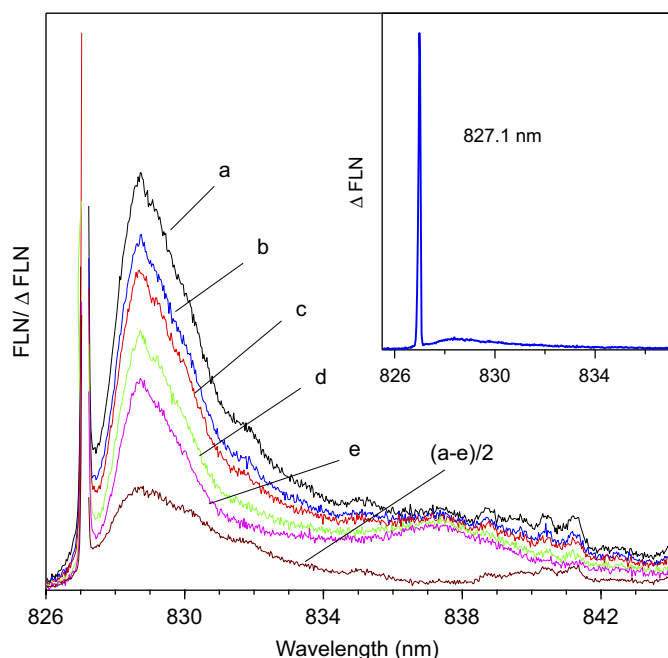


Fig. 4. FLN and Δ FLN spectra of the FMO complex close to the 0–0 origin line and its phonon sideband. The same excitation/burning laser wavelength at 827.1 nm is used. The five top FLN spectra (a through e) correspond to the burn/measuring fluences of 1.6 (no extra burning), 4.8, 8.0, 28, and 126 mJ/cm². The lowest curve shows a difference between the initial and the final fluorescence spectra, the Δ FLN spectrum. An overview Δ FLN spectrum is shown in the inset.

absorption band maximum, see inset in Fig. 3) was used throughout interchanged burning and measuring cycles. To reduce hole burning during the recordings, all the spectra were measured with the minimal excitation fluence of 1.6 mJ/cm², which still ensures reasonable signal-to-noise ratio. The lowest curve in Fig. 4 shows the difference between the first and the last spectra in the series, i.e. the Δ FLN spectrum obtained at total fluence of 126 mJ/cm². The inset in Fig. 4 displays the complete Δ FLN spectrum of the pure electronic band with sharp 0–0 origin line and its accompanying PSB. The shape of the ZPL line is not fully resolved due to the limited detection resolution of ~ 1.5 cm⁻¹ used. The actual homogeneously broadened 0–0 line is much narrower and correspondingly more intense than appears in this figure.

Fig. 5 shows the entire Δ FLN spectrum that is recorded using the same excitation as in Fig. 4, but considerably smaller burn fluence of 28 mJ/cm². The lower energy region of the spectrum, corresponding to higher vibrational frequencies, is shown in the inset. The ZPL is cut off at $\sim 4\%$ of its peak intensity. In total 62 vibrational modes, including the low-frequency ones of 46, 68 and 117 cm⁻¹ on top of the continuous phonon spectrum (see Section 5), could be identified in this spectrum.

Fig. 6 represents the Δ FLN spectra measured using excitation wavelengths all over the 826 nm absorption band (the corresponding excitation positions are shown by

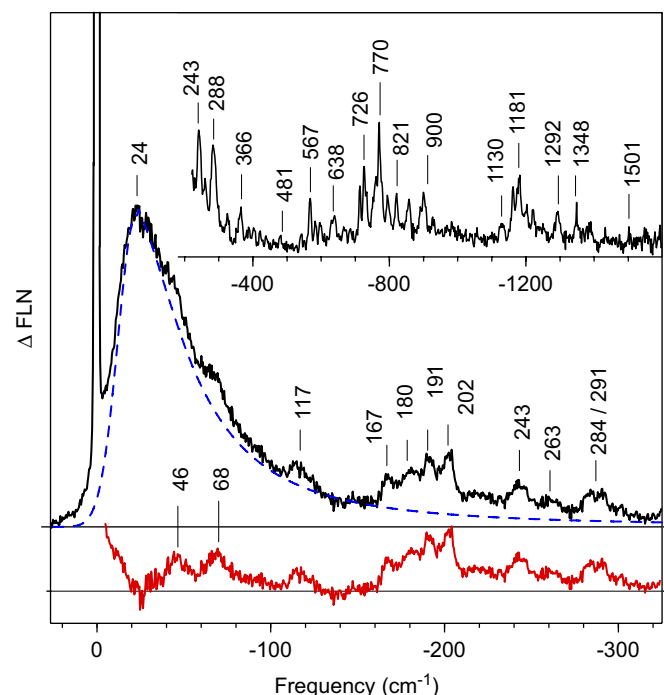


Fig. 5. The full-length Δ FLN spectrum of the FMO complex at 827.1 nm excitation and burn fluence of 28 mJ/cm². The lower energy region of the spectrum, corresponding to higher vibrational frequencies, is shown in the inset. The ZPL is cut off at $\sim 4\%$ of its peak intensity. The calculated phonon contribution is shown by dashed line. The curve at the bottom represents the difference between experimental and calculated spectra. The numbers indicate vibrational frequencies in wave numbers.

vertical lines on the inset in Fig. 3). The top curves in each frame are the initial FLN spectra, recorded before intentional hole burning of the sample (the pre-burn spectra). The blue dashed curves below are the post-burn FLN spectra obtained after burning with high-burn fluence of 8 (frame (a)) or 126 mJ/cm² (frames (b–f)). Differences between the pre-burn and post-burn fluorescence signals, the Δ FLN spectra, are presented with red curves. Most of the variations of the Δ FLN spectra on excitation wavelength evident on Fig. 6 are quite expected from the analysis provided in Section 2. Yet, the conservative spectral band seen around 836–839 nm does not readily fit into the presented model. Residuals of reaction center complexes would be consistent with both the spectral characteristics and high resistance to hole burning of the band [2]. It remains to be seen whether this interpretation is feasible.

Some deterioration of the vibronic structure at large exposures can be noticed on Fig. 4, the well-known hole-burning saturation effect (see Refs. [15,17–19] for reviews). As the rates of burning of ZPL and PSB are different, the phonon band in the Δ FLN spectrum may appear more prominent than it is in the actual homogeneous spectrum. To assess the error made by this effect on electron–phonon coupling factors, the burn fluence dependence of the ZPL and PSB intensities in the Δ FLN spectrum was studied. As can be seen from Fig. 7, at lower exposures smaller

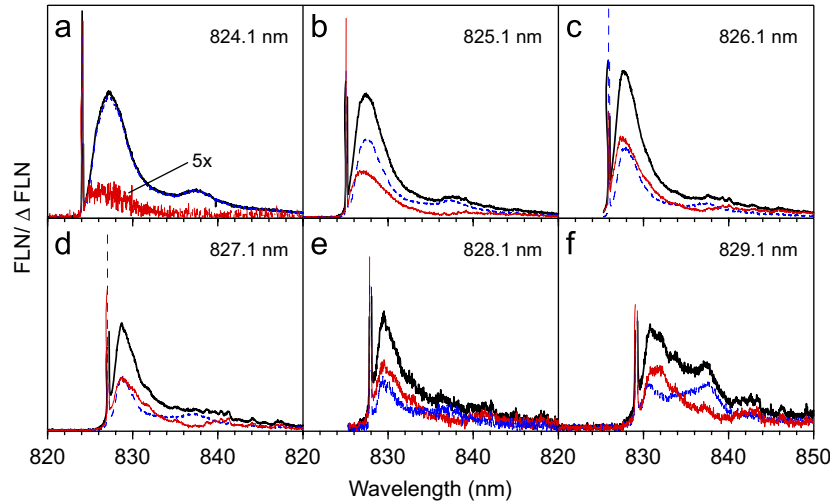


Fig. 6. Δ FLN spectra (red curves) of the FMO complex excited at different wavelengths indicated. Black solid and blue dashed curves show the corresponding pre- and post-burn FLN spectra. Burn fluences of 8 (a) and 126 (b–f) mJ/cm^2 were used.

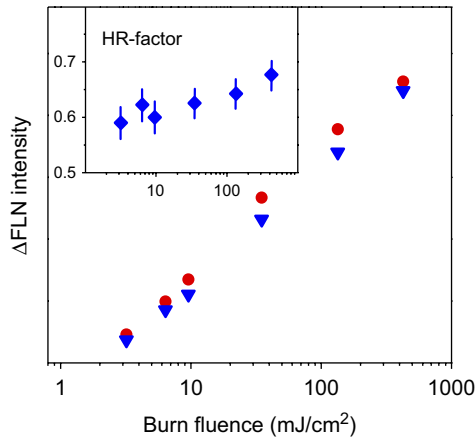


Fig. 7. Burn fluence dependence of the ZPL (rings) and PSB (triangles) intensities in the Δ FLN spectrum excited/burnt at 871.1 nm. The inset shows electron–phonon coupling strength values calculated from these data. No correction on the non-resonant excitation effects, as in Fig. 8, was carried out.

apparent HR factors are acquired. (Presented in this figure are the raw S_{ph} data, not corrected for PSB components, explaining their relatively large numerical value.) Despite decreasing gradient of this dependence at lower fluences, it is clear that extrapolation to zero is necessary for obtaining the correct coupling values. However, the data of Fig. 7 also show that errors owing to other, yet to be identified, factors overcome the ones due to saturation effects already at about 10–30 mJ/cm^2 . This range of fluences was typically used in our experiments.

4. Data analysis and discussion

Evaluation of the electron–phonon and vibronic coupling strengths is not straightforward even by knowing the intensity of the 0–0 origin line. As it was explained in

Section 2, this is mainly because experimental spectra contain different PSB contributions, from which only the real-PSB component is relevant. Numerical simulations based on some model are, therefore, necessary to disentangle the different contributions. Below, we shall describe such a procedure within the framework of the band shape model developed in Ref. [21].

The homogeneously broadened band shapes of an impurity molecule in the low-temperature limit may be written down as

$$L(\omega) = e^{-S} l_0(\omega - \Omega) + \sum_{R=1}^{\infty} e^{-S} \frac{S^R}{R!} l_R(\omega - \Omega \mp R\omega_m), \quad (2)$$

where the minus and plus signs correspond to the absorption and fluorescence spectra, respectively; $l_0(\omega - \Omega)$ is the normalized shape function of the zero phonon and zero vibration origin line at frequency Ω and ω_m represents the phonon and vibration peaks in the one-quantum profile, $l_1(\omega - \Omega \mp \omega_m)$. The latter is a product of two frequency-dependent distributions, the density of vibrating states and the electron–vibrational coupling strength. The sideband of the origin line is formed as a sum of all R -quanta ($R \geq 1$) contributions, obtained by R times folding of the normalized line shape, $l_1(\omega - \Omega \mp \omega_m)$, with itself. The already familiar Franck–Condon factors determine relative intensity of the R th order contributions into the sideband. For comparison with the experiment, the homogeneous spectrum of Eq. (2) should be convoluted with the inhomogeneous site distribution function, SDF, and multiplied by ω or ω^3 in the case of absorption and fluorescence spectra, respectively [5].

As defined above, the one-quantum profile, $l_1(\omega - \Omega \mp \omega_m)$, takes care of all vibrations coupled to the electronic transition [22]. No distinction is made between localized and delocalized vibrational degrees of freedom. Below, a more conventional [21], yet physically,

well-justified view will be taken that separately deliberates the phonon and intramolecular local modes. We, therefore, assume that the whole continual part of the Δ FLN spectrum (see Fig. 5) is due to electron–phonon coupling. Recovering of the homogeneous origin band is needed in order to evaluate the S_{ph} value from the 0–0 line and PSB intensities, I_{ZPL} and I_{PSB} , respectively [5]

$$\exp(-S_{\text{ph}}) = \frac{I_{\text{ZPL}}}{I_{\text{ZPL}} + I_{\text{PSB}}}. \quad (3)$$

In the experimental section, it was explained that iterative simulations of the Δ FLN spectra might help finding the homogeneous spectrum. Variables of these simulations are the peak position and width of SDF, S_{ph} , and the one-phonon profile, $I_1(\omega - \Omega + \omega_m)$. Since the SDF can be determined in separate hole-burning experiment, we are left with S_{ph} and $I_1(\omega - \Omega + \omega_m)$ as the only model parameters. This is still too much of a freedom for a unique solution, unless additional limitations for the shape of $I_1(\omega - \Omega + \omega_m)$ are introduced. A reasonable guess, looking at the PSB on Fig. 5, would be a single-peak skewed form. For computational ease, the one-phonon profile was constructed from Gaussian and Lorentzian parts representing the low- and high-frequency sides of the distribution, respectively. By modifying S_{ph} and halfwidths of the Gaussian and Lorentzian sides self-consistently, we then aimed at best fit of the measured and calculated spectra.

Fig. 5, and in more detail Fig. 8, represents a result of such simulation at 827.1 nm excitation. The fit of the experimental spectrum is very good, leaving of course out those regions where the low-frequency (quasi) localized (see below) modes overlap with the PSB. As can be seen from

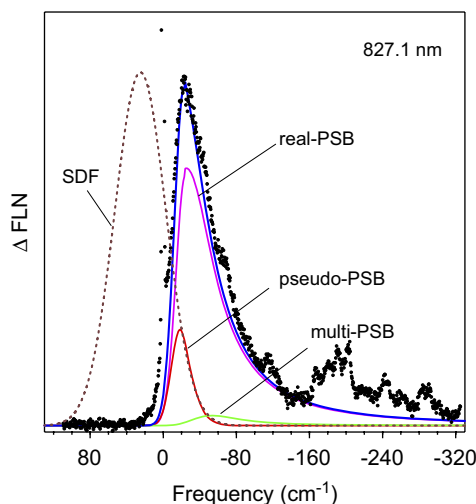


Fig. 8. The experimental (dots, burn fluence of 28 mJ/cm² at 827.1 nm) and calculated (blue solid line) Δ FLN spectra of the FMO complex. The ZPL is dropped out for clarity. The three contributions to the calculated spectrum, accounting for only the phonon part, are labeled as real-PSB, pseudo-PSB, and multi-PSB. The model parameters used are $S_{\text{ph}} = 0.52$, $\omega_m = 22 \text{ cm}^{-1}$, $\Delta = 11 \text{ cm}^{-1}$ (halfwidth of the Gaussian side), and $\Gamma = 30 \text{ cm}^{-1}$ (halfwidth of the Lorentzian side). The SDF distribution (dashed line) peaking at $12,105 \text{ cm}^{-1}$ is also shown.

Fig. 8, at this particular excitation, the main contribution to the phonon spectrum is given by the real-PSB, the high-frequency tail of the pseudo-PSB signal being effectively suppressed by the limited width of SDF. The SDF shown on this figure and also in the inset of Fig. 3 is from Ref. [23]. We only checked that it is very similar to the one in our case. The best-fit effective HR factor for phonons, calculated according to Eq. (3) using the areas of the ZPL and real-PSB, equals to $S_{\text{ph}} = 0.52 \pm 0.06$. The uncertainty of this number accounts for all experimental errors we could think about, one of the most serious obstacle being the residual level of resonantly scattered light in the Δ FLN spectrum relatively to the fluorescence intensity. Two principal factors that control this level are (i) the scattered light intensity in the FLN spectrum and (ii) the stability of the experimental set-up. The former aspect depends primarily on the sample quality, while the second one, on stability of the lasers. In our experiments, the absolute scattered light level was typically kept of the order of I_{ZPL} . The reproducibility of the repeated measurements on different samples was about 10%. It can be estimated that the error ± 0.06 at the mean value of S_{ph} equal to 0.52 corresponds to the variance in I_{ZPL} of $\sim 15\%$.

In accordance with single-molecule spectroscopy data [24,25], spectral variances of the effective HR factors over the inhomogeneous ensemble of molecules are expected. Fig. 9 shows a dependence of S_{ph} on burn wavelength across the inhomogeneously broadened 826 nm absorption band. The phonon coupling strength appears nearly constant between 0.6 and 0.7 at short and center wavelengths, but decreases rapidly at the long-wavelength slope of the absorption band. Energy transfer from the directly excited pigments or exciton states to the lower-energy ones could be a cause of this wavelength dependence. Various indications for such energy transfer have been reported [23,26,27]. The actual, undisturbed S_{ph} values may, therefore, emerge only at the very red edge of the absorption spectrum where the energy transfer effects are minor (see, however, in Summary).

We proceed with evaluation of the vibronic HR factors. The intensity of the vibronic spectrum is determined by

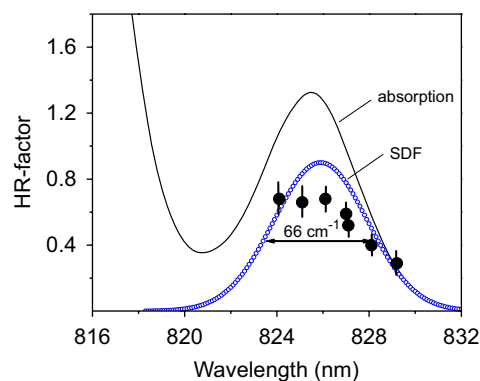


Fig. 9. Dependence of S_{ph} (closed rings) on excitation wavelength. The absorption spectrum and SDF are drawn for reference.

subtracting the above calculated phonon spectrum from the overall measured Δ FLN spectrum (see Fig. 5). The result for the lower 310 cm^{-1} energy range is shown at the bottom of Fig. 5. The implication of this procedure is that the phonon and vibronic parts of the spectrum are additive. Using the relationship $|\langle 1|0\rangle|^2/|\langle 0|0\rangle|^2 = S$ introduced in Section 1, we then assessed the HR factors for individual modes as the relative intensities of particular modes over the intensity of the whole 0–0 band (including its ZPL and phonon parts). This is justified by the smallness of S_j that reduces the sum in Eq. (3) to a single one-vibration term of weight $S_j \exp(-S_j) \approx S_j$. In multimode case, the number of such terms is of course equal to the number of local modes.

The individual vibronic HR factors evaluated are listed in Table 1. They sum up to $S_{\text{vib}} = \sum_j S_j = 0.31$. A slightly larger figure of $S_{\text{vib}} = 0.38$ was found when the intensity of the entire vibronic spectrum in the range of $30\text{--}1600\text{ cm}^{-1}$ was summed up. A small number of weak modes may have been overlooked by the individual mode accounting approach.

Assuming insignificant dependence of S_{vib} on excitation wavelength, the total coupling strength, $S_{\text{total}} = S_{\text{ph}} + S_{\text{vib}}$, measured at the red edge of FMO absorption spectrum can be estimated as $S_{\text{total}} \approx 0.29 + 0.38 = 0.67 \pm 0.10$. The strengths of electron–phonon and vibronic couplings reported by different selective spectroscopy approaches are compared in Table 2. It comes as a little surprise that all the techniques have reached similar S_{ph} values of about 0.3, despite various environments of the BChl molecules in different experiments. The present S_{total} and S_{vib} values agree with those in Refs. [8,25] within the experimental

Table 2

Comparison of electron–phonon and vibronic coupling strength values determined by different methods for the Q_y electronic transition of the bacteriochlorophyll *a* molecule in various glassy host surroundings

Method	S_{ph}	S_{vib}	S_{total}	Comments
Δ FLN	0.29 ± 0.03	0.38 ± 0.07	0.67 ± 0.10	FMO from <i>Cb. tepidum</i> measured at 829.1 nm, this work
FLN	0.3	0.15	0.45 ± 0.05	FMO from <i>P. aestuarii</i> , Ref. [9]
SHB	0.3	0.45	0.75 ± 0.10	Glycerol/water/LDAO, Ref. [8]

uncertainty. The significantly smaller S_{total} found in Ref. [9] was determined from the temperature broadening of the absorption spectrum. This method in principle deals only with thermally activated modes, underestimating the role of high-frequency vibrations.

5. Summary and concluding remarks

There are a few merits of Δ FLN raising it in a very favorable position compared with other two spectrally selective techniques considered in this work, SHB and FLN. The main advantage of Δ FLN relative to the common FLN is that ZPLs resonant with the excitation laser can be accurately measured. Compared with SHB, on the other hand, in Δ FLN the limited width of the inhomogeneous SDF and uncertainties with the hole-burning products cause far less difficulties. Common for the all three techniques, however, is that the low-frequency sidebands of the narrow resonant line consist from several contributions. Only one of these contributions, referred to above as real-PSB, is associated with the molecules excited directly through their ZPLs. Others are due to off-resonant molecules that absorb via their broad PSB. Neglect of these latter contributions results in an overestimation of electron–phonon (and concomitant underestimation of vibronic) couplings. Because the Δ FLN spectrum provides not only the PSB and vibrational structures but it also accurately measures ZPL, the real-PSB can with relative ease be extracted by computer simulation of the spectrum. This way over 60 vibrational modes between 45 and 1600 cm^{-1} that distinguish the coupled BChl guest–FMO protein host system were detected and their coupling strengths to the lowest-energy electronic transition determined at 4.5 K.

The effective HR factor for the phonons, equal to about 0.3 at the long-wavelength tail of the absorption spectrum, was found clearly wavelength dependent. As shown in Ref. [28], a coupling between the BChl *a* molecules is likely responsible for this wavelength dependence.

Despite the precise origin of the three lowest-frequency vibrations in Fig. 5 (46 , 68 , and 117 cm^{-1}), they should be classified as pseudo-local rather than local modest, because of strong overlap with the continuous phonon spectrum and of relatively large breadth. In the *P. aestuarii* matrix,

Table 1

Frequencies, ν_j ($\pm 2\text{ cm}^{-1}$), and relative intensities, S_j (± 0.0005), of vibrational modes in the Δ FLN spectrum of BChl molecules in FMO complexes from *Cb. tepidum* at 4.5 K

ν_j (cm^{-1})	S_j	ν (cm^{-1})	S_j	ν (cm^{-1})	S_j
46	0.011	597	0.004	1004	0.001
68	0.011	630	0.003	1037	0.002
117	0.009	638	0.006	1058	0.002
167	0.009	665	0.004	1094	0.001
180	0.010	684	0.003	1104	0.001
191	0.011	713	0.007	1123	0.003
202	0.011	726	0.010	1130	0.003
243	0.012	731	0.005	1162	0.009
263	0.003	750	0.004	1175	0.007
284	0.008	761	0.009	1181	0.010
291	0.008	770	0.018	1201	0.003
327	0.003	795	0.007	1220	0.005
366	0.006	821	0.006	1283	0.002
385	0.002	856	0.007	1292	0.004
404	0.002	891	0.003	1348	0.007
423	0.002	900	0.004	1367	0.002
440	0.001	924	0.001	1386	0.004
481	0.002	929	0.001	1431	0.002
541	0.004	946	0.002	1503	0.003
568	0.007	966	0.002	1545	0.003
582	0.004	984	0.003	–	–

Highlighted are the S_j values equal or larger than 0.005.

these modes appear at somewhat different frequencies of 36, 70, and 117 cm^{-1} , respectively [9].

The approach used for the analysis is based on the suggestion of linear electron–phonon and vibronic couplings. A legitimate question, why quadratic coupling terms have been neglected, may rise. The answer is based on the following arguments. The quadratic coupling assumes changes of phonon and vibrational frequencies as well as mixing of their coordinates at electronic transition. Experimentally, the coupling reveals itself in a breakdown of mirror symmetry of conjugate absorption and emission spectra corresponding to this transition. Only small (within $\sim 1\%$ or less) variations of BChl vibrational frequencies have been observed in glassy matrices [7] with no evidence of mode mixing. Although the quadratic coupling is able to modify the shape of the ZPL sidebands, its influence on their intensity is rather weak [29]. The two physical phenomena that are known to be mostly responsible for the strength of the PSBs are the linear coupling and the dependence of the electronic transition matrix element on vibrational coordinates (also known as non-Condon or Herzberg–Teller coupling). As long as the non-Condon effects do not play any role in our spectra, for which there are no indications, the HR factors are valid characteristics of the electron–phonon and vibronic coupling strengths present.

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References

- [1] M. Lax, *J. Chem. Phys.* 20 (1952) 1752.
- [2] M. Miller, R.P. Cox, J.M. Olson, *Photosynth. Res.* 41 (1994) 97.
- [3] R.M. Pearlstein, in: H. Scheer (Ed.), *Chlorophylls*, CRC, Boca Raton, FL, 1991, p. 1047.
- [4] R.J.W. Louwe, J. Vrieze, A.J. Hoff, T.J. Aartsma, *J. Phys. Chem. B* 101 (1997) 11280.
- [5] K.K. Rebane, *Impurity Spectra of Solids*, Plenum Press, New York, 1970.
- [6] R.A. Avarmaa, K.K. Rebane, *Spectrochim. Acta A* 41 (1985) 1365.
- [7] I. Renge, K. Mauring, R.A. Avarmaa, *J. Lumin.* 37 (1987) 207.
- [8] V. Zazubovich, I. Tibe, G.J. Small, *J. Phys. Chem. B* 105 (2001) 12410.
- [9] M. Wendling, T. Pullerits, M.A. Przyjalowski, S.I. Vulto, T.J. Aartsma, R. van Grondelle, H. van Amerongen, *J. Phys. Chem. B* 104 (2000) 5825.
- [10] J. Pieper, R. Schödel, K.-D. Irrgang, J. Voigt, G. Renger, *J. Phys. Chem. B* 105 (2001) 7115.
- [11] M. Rätsep, A. Freiberg, *Chem. Phys. Lett.* 377 (2003) 371.
- [12] K. Timpmann, M. Rätsep, C.N. Hunter, A. Freiberg, *J. Phys. Chem. B* 108 (2004) 10581.
- [13] A.A. Gorokhovskii, R.K. Kaarli, L.A. Rebane, *JETP Lett.* 20 (1974) 216.
- [14] B.M. Kharlamov, R.I. Personov, L.A. Bykovskaya, *Opt. Commun.* 12 (1974) 191.
- [15] O. Sild, K. Haller (Eds.), *Zero-Phonon Lines and Spectral Hole Burning in Spectroscopy and Photochemistry*, Springer, Berlin, Heidelberg, 1988.
- [16] A. Szabo, *Phys. Rev. Lett.* 25 (1970) 924.
- [17] R.V. Jaaniso, R.A. Avarmaa, *Sov. J. Appl. Spectrosc. (USA)* 44 (1986) 365.
- [18] J. Fünfschilling, D. Glatz, I. Zschokke-Gränacher, *J. Lumin.* 36 (1986) 85.
- [19] R. Jankowiak, in: C. Gooijer, F. Ariele, J.W. Hofstra (Eds.), *Shpol'skii Spectroscopy and Other Site-Selective Methods*, Wiley, New York, 2000, p. 235.
- [20] M. Rätsep, C.N. Hunter, J.D. Olsen, A. Freiberg, *Photosynth. Res.* 86 (2005) 37.
- [21] J.M. Hayes, P.A. Lyle, G.J. Small, *J. Phys. Chem.* 98 (1994) 7337.
- [22] T. Pullerits, R. Monshouwer, F. van Mourik, R. van Grondelle, *Chem. Phys.* 194 (1995) 395.
- [23] M. Rätsep, R.E. Blankenship, G.J. Small, *J. Phys. Chem. B* 103 (1999) 5736.
- [24] J.G. Müller, M. Anni, U. Scherf, J.M. Lupton, J. Feldmann, *Phys. Rev. B* 70 (2004) 035205.
- [25] C. Hofmann, H. Michel, M. van Heel, J. Köhler, *Phys. Rev. Lett.* 94 (2005) 195501.
- [26] R.J.W. Louwe, T.J. Aartsma, *J. Phys. Chem. B* 101 (1997) 7221.
- [27] A. Freiberg, S. Lin, K. Timpmann, R.E. Blankenship, *J. Phys. Chem. B* 101 (1997) 7211.
- [28] M. Rätsep, A. Freiberg, *Chem. Phys. Lett.* 344 (2007) 306.
- [29] I.S. Osad'ko, in: V.M. Agranovich, R.M. Hochstrasser (Eds.), *Spectroscopy and Excitation Dynamics of Condensed Molecular Systems*, North-Holland, Amsterdam, 1983, p. 438.